

Marin Tools

CLI Reference Manual

All tools live in `tools/` and share a consistent `--flag` interface. They print a banner, show results on the terminal, and may open a GUI window (chart, price tracker, game). Tools are launched by `marin_fier.py` when Marin detects the matching intent, or can be run directly from the command line.

1. stock.py - Stock Price + 30-Day Chart

\$ python3 tools/stock.py --ticker AAPL

Fetch live price and 30-day chart for a stock ticker. Opens a matplotlib chart window and prints a price table to terminal with dates and closing prices.

Example: `python3 tools/stock.py --ticker META`

Note: Use --company "Company Name" to auto-resolve the ticker via Yahoo search.

2. crypto.py - Live Crypto Price Tracker

\$ python3 tools/crypto.py --coin ethereum

Open a tkinter window showing live USD price of the given cryptocurrency, updating every second via CoinGecko API.

Example: `python3 tools/crypto.py --coin bitcoin`

Note: Defaults to 'bitcoin' if --coin is omitted.

3. news.py - Open News Website

\$ python3 tools/news.py --source BBC

Open a news website in the default browser. Supported sources: BBC, AlJazeera, NDTV.

Example: `python3 tools/news.py --source AlJazeera`

Note: Defaults to BBC if --source is omitted.

4. alarm.py - Set an Alarm

\$ python3 tools/alarm.py --time "05:00"

Set a clock alarm. Accepts HH:MM or HH:MM AM/PM. Runs in the background (forks to child process) and beeps at the target time using `alarm.wav` (via `pygame mixer`).

Example: `python3 tools/alarm.py --time "7:30 AM"`

Note: The process daemonizes so your terminal stays free.

5. timer.py - Countdown Timer

\$ python3 tools/timer.py --duration 300

Start a countdown timer. Duration is in seconds. Beeps when time runs up using alarm.wav.

Example: `python3 tools/timer.py --duration 90`

Note: 300 seconds = 5 minutes, 90 seconds = 1:30.

6. translate.py - Dictionary Translator

\$ python3 tools/translate.py --text "I love you" --to bn

Translate English text to a target language. Prints the result to terminal and speaks it aloud via TTS. Accepts language names (bangla, french) or codes (bn, fr).

Example: `python3 tools/translate.py --text "Good morning" --to french`

Note: Uses deep_translator (GoogleTranslate) with googletrans fallback.

7. image.py - Screenshot to Stencil Art

\$ python3 tools/image.py --prompt "Starry Night"

Capture a screen region (uses maim) and convert it to a black-and-white stencil image displayed in a window.

Example: `python3 tools/image.py --prompt "My Desktop"`

Note: Requires maim: `sudo apt install maim`

8. Games - TicTacToe / Connect4 / WordGame

python3 tools/tictactoe.py

Classic Tic Tac Toe game.

python3 tools/connect4.py [--two]

Connect Four. --two for 2-player mode.

python3 tools/wordgame.py

Word scramble / word game.

9. Other Tools

python3 tools/draw_me.py

Marin Tools - CLI Reference

Take a webcam photo and render it as turtle drawing.

```
python3 tools/email_tool.py
```

Interactive email composer (GUI).

```
python3 tools/bangla.py
```

Bangla language utility.

```
python3 tools/vpa.py
```

VPA utility tool.

```
python3 tools/draw.py
```

Drawing utility.

10. Integration with Marin

marin_fier.py is the two-stage intent classifier that maps user messages to tool calls. Stage 1 uses regex for obvious patterns ("stock", "alarm", "ticker TSLA"). Stage 2 uses qwen2.5:0.5b with LangChain StructuredTool bindings for fuzzy/ambiguous requests. The execute_tool() function runs the matching tool via subprocess.Popen in a detached session, so each tool runs as its own independent process.

```
Intent detection order:  crypto > stock > bare ticker > alarm >  
timer > news > email > tictactoe > connect4 > wordgame >  
draw_me > screenshot > run_command > qwen-llm fallback
```